

Table 6.2 Estimated Parameters

Index	EWMA	GARCH		
	λ	Ω	α	β
SP500	0.8808	3.42E-06	0.1519	0.8420
R2000	0.8758	9.92E-06	0.1614	0.8222

evaluation statistics that can be used to further critique the accuracy of these models. Our usage of these aforementioned performance statistics is to provide a point of comparison across techniques. These procedures are:

Root Mean Square Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum (\hat{\sigma}_t - \sigma_t)^2} \quad (6.28)$$

The RMSE is simply the difference squared between the estimated volatility $\hat{\sigma}_t$ and realized volatility. Realized volatility was measured as the square root of squared return (e.g., absolute value of return), that is, $\sigma_t = \sqrt{y_t^2}$. This follows along the lines of the more traditional statistical tests such as minimizing sum of squares used in regression analysis.

Root Mean Z-Score Squared Error (RMZSE)

$$RMZSE = \sqrt{\frac{1}{n} \sum \left(\frac{y_t^2}{\hat{\sigma}_t^2} - 1 \right)^2} \quad (6.29)$$

The RMZSE is a measurement of the squared difference between our test statistic z and one.

This test is derived as follows. Let, $z = \frac{y - \mu}{\sigma}$. Then we have $E[z] = 0$ and $Var[z] = 1$.

Since we have $y_t \sim N(0, \hat{\sigma}_t^2)$, our test statistic z can be written as $z_t = \frac{y_t}{\hat{\sigma}_t}$. And the variance of z is simply, $Var[z_t] = \frac{y_t^2}{\hat{\sigma}_t^2} = 1$. The root mean z-score squared error is then a test of how close the test statistic is to its theoretical value.